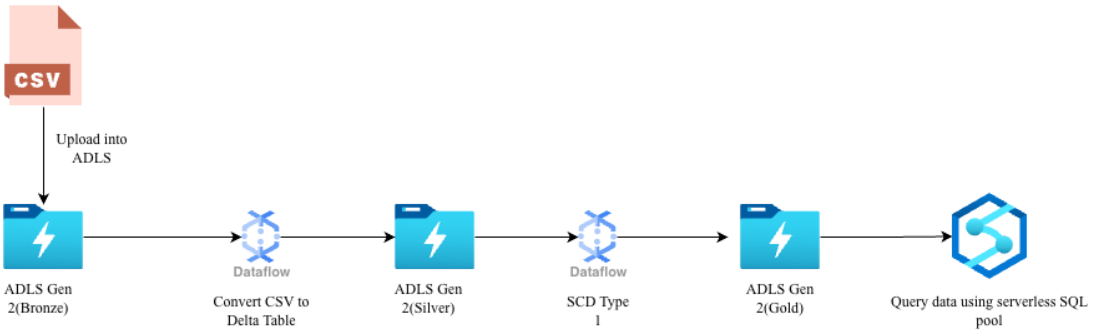
Data Warehousing with Azure Synapse Analytics

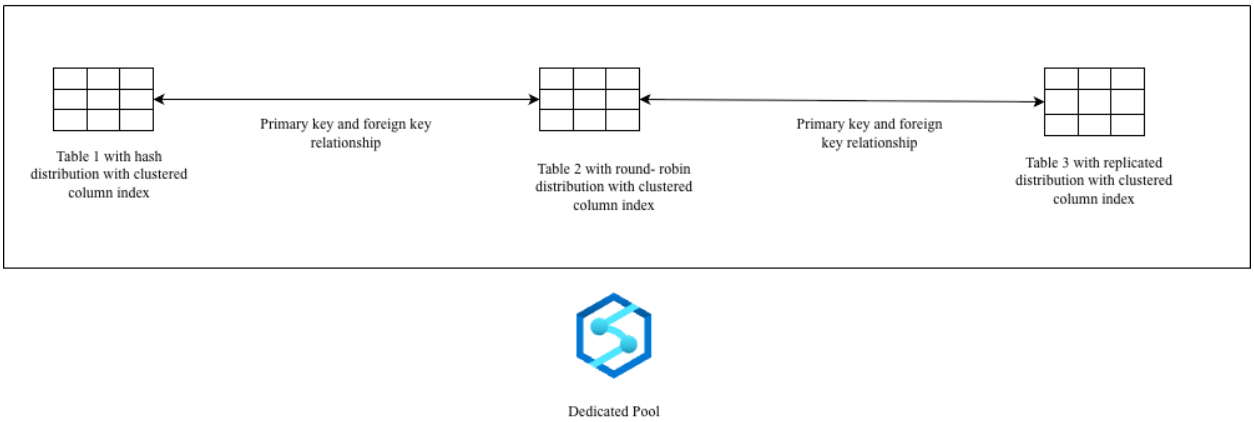
- **Rithika Gopi**

Objective:

To build a simple data warehouse solution using Azure Synapse Analytics**,** ADLS Gen2, and Power BI by ingesting raw sales data, transforming it into Silver and Gold layers, implementing SCD Type 1 for cleaned business data, querying the final data using serverless SQL pool, and designing dedicated SQL pool tables with different distribution methods and clustered columnstore indexes for reporting and analytics.

Architecture:





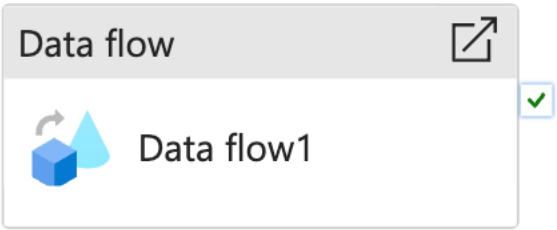
Steps:

Step 1: Add your dataset to your ADLS Gen 2 bronze container.

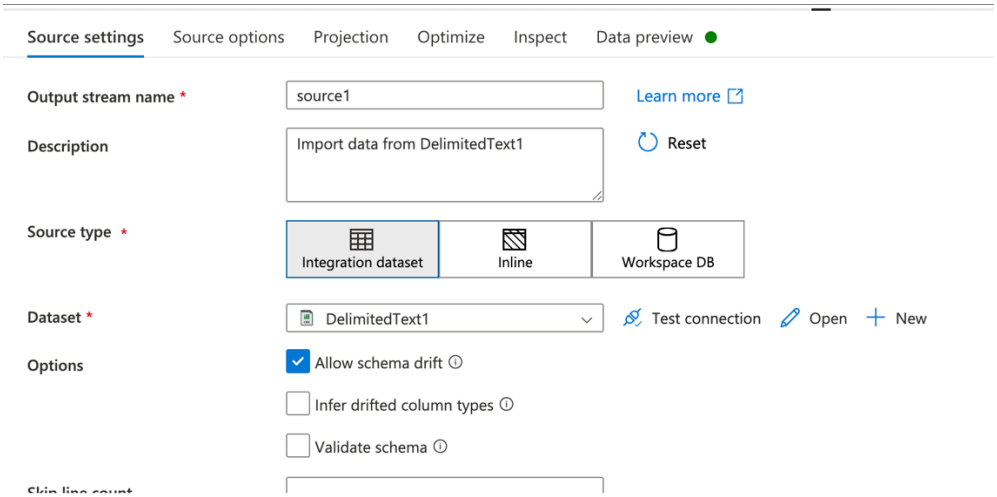
Dataset: https://www.kaggle.com/datasets/vinothkannaece/sales-dataset

Step 2: Open your synapse, go to integrate and create a new pipeline.

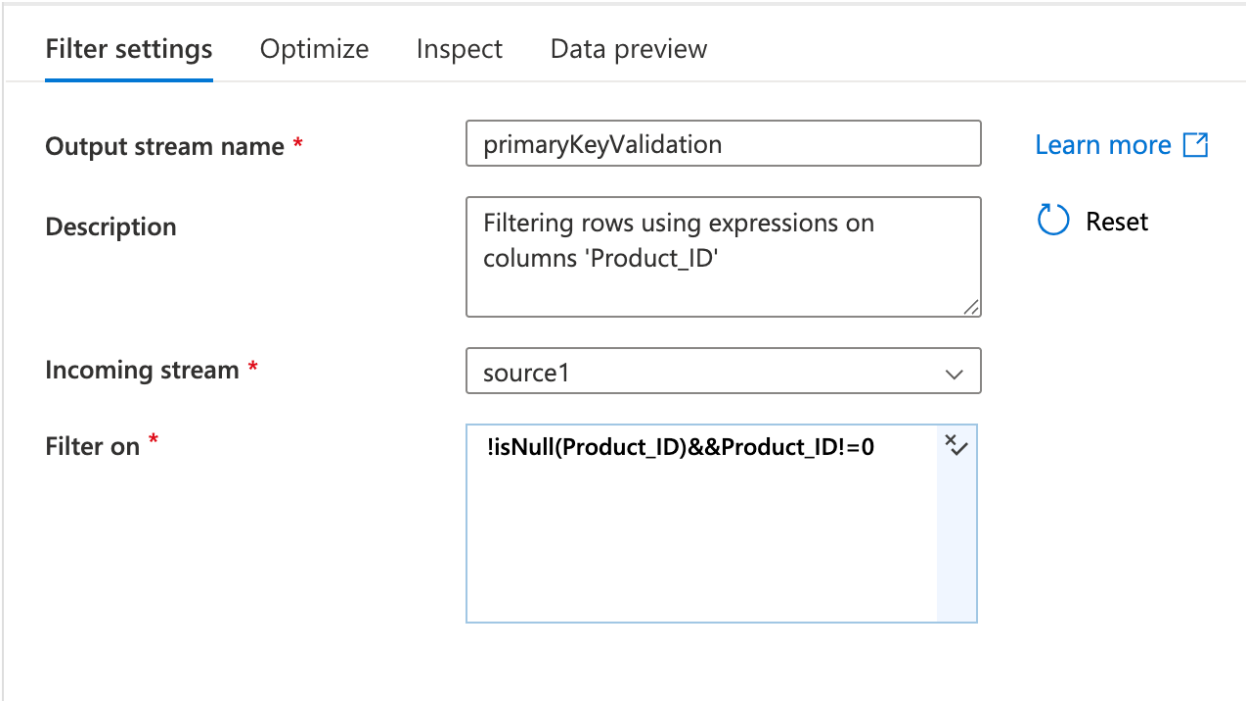
Step 3: Add a dataflow activity



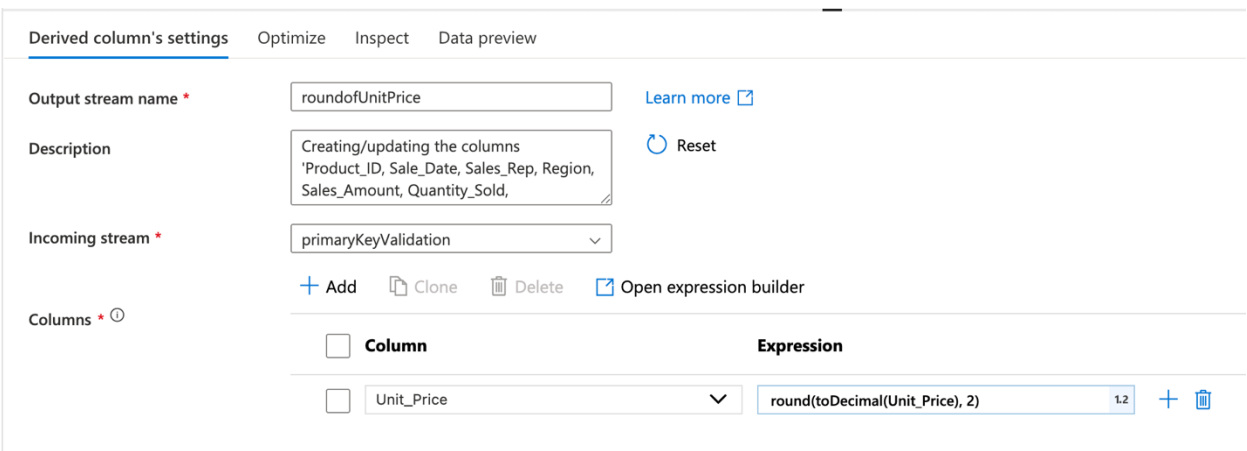
Step 4: Inside your dataflow give source as your csv that you saved in bronze



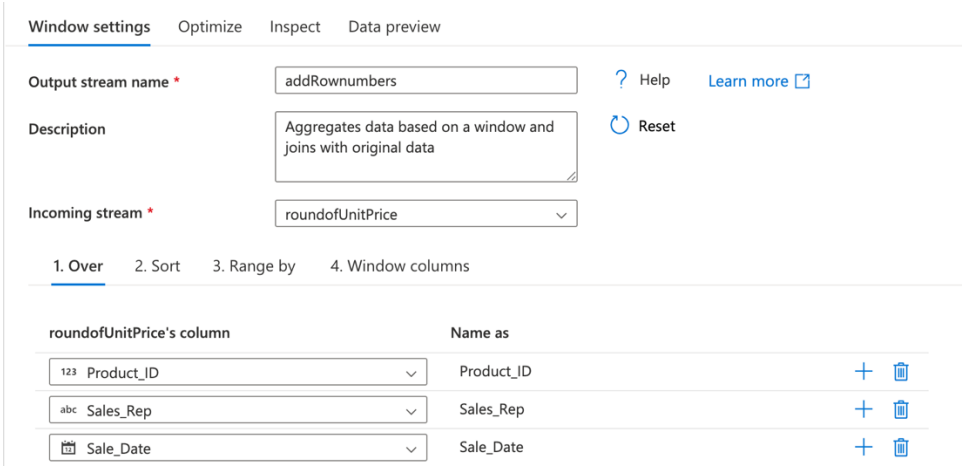
Step 5: Then add a filter activity to validate our primary key

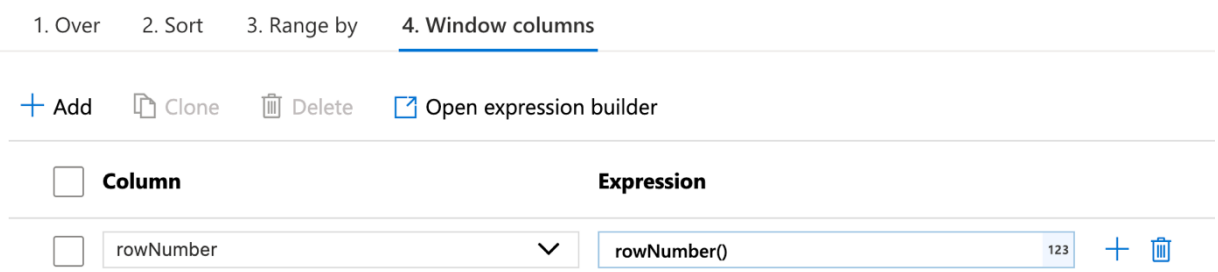


Step 6: Then add a derived column to round off the unit_price column to 2 digits.

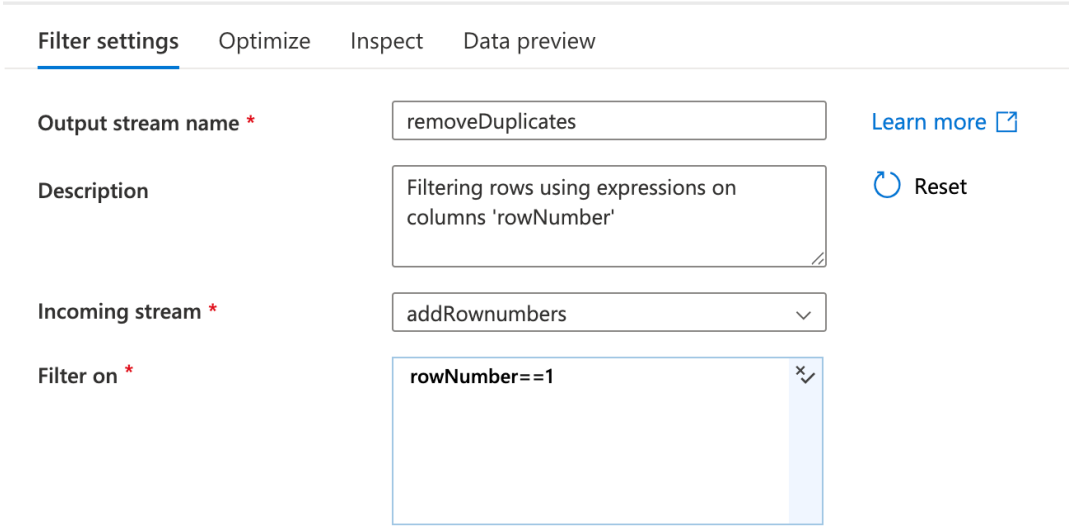


Step 7: Then add a windows activity to add row number column to remove duplicates.

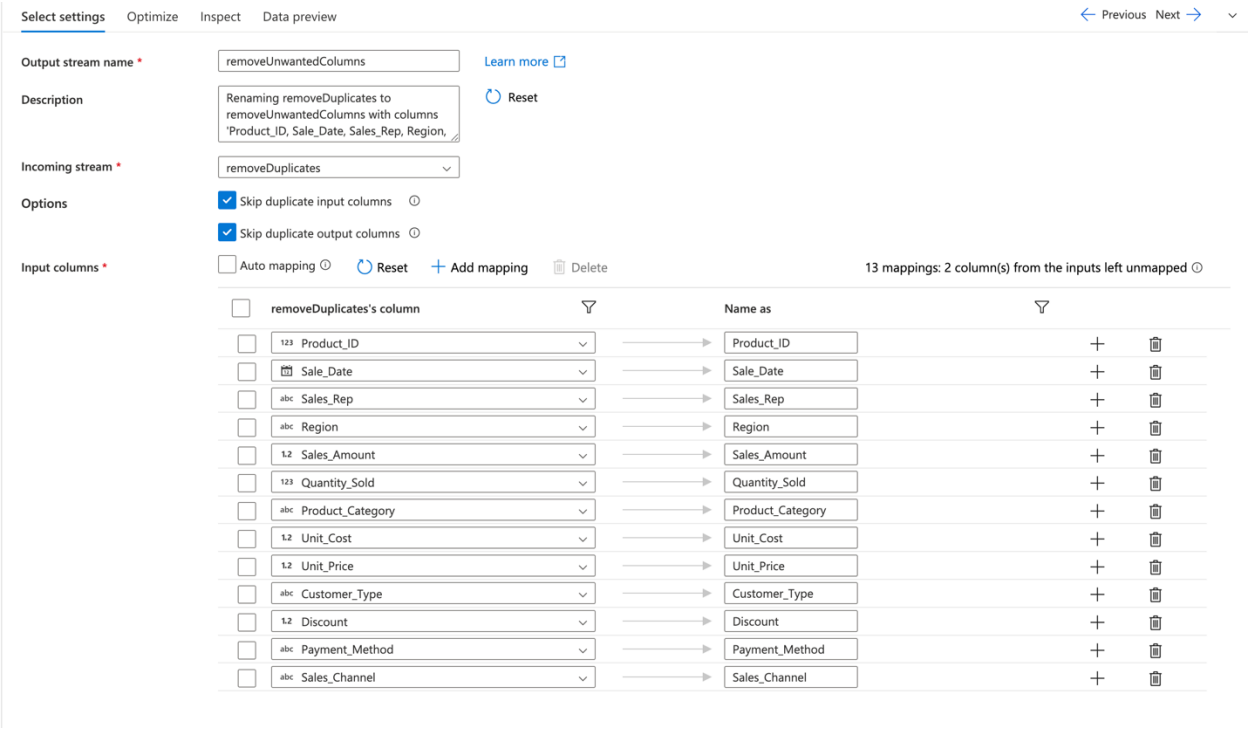




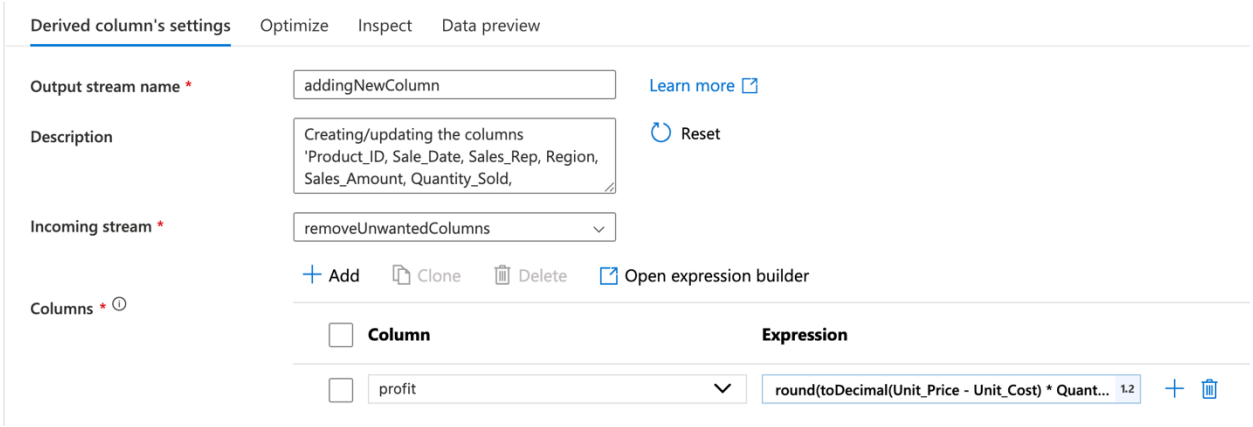
Step 8: Then add a filter to remove all the duplicates



Step 9: Then add a select to remove unwanted columns

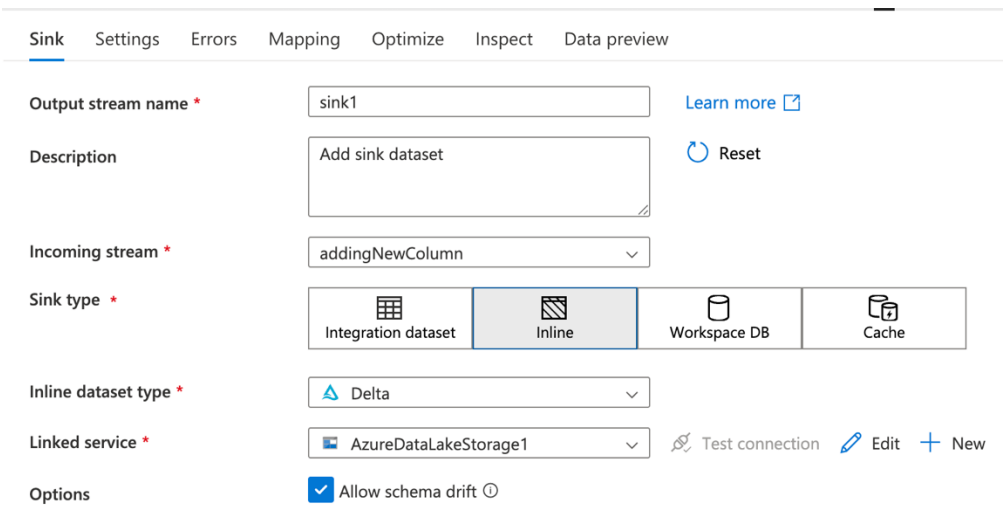


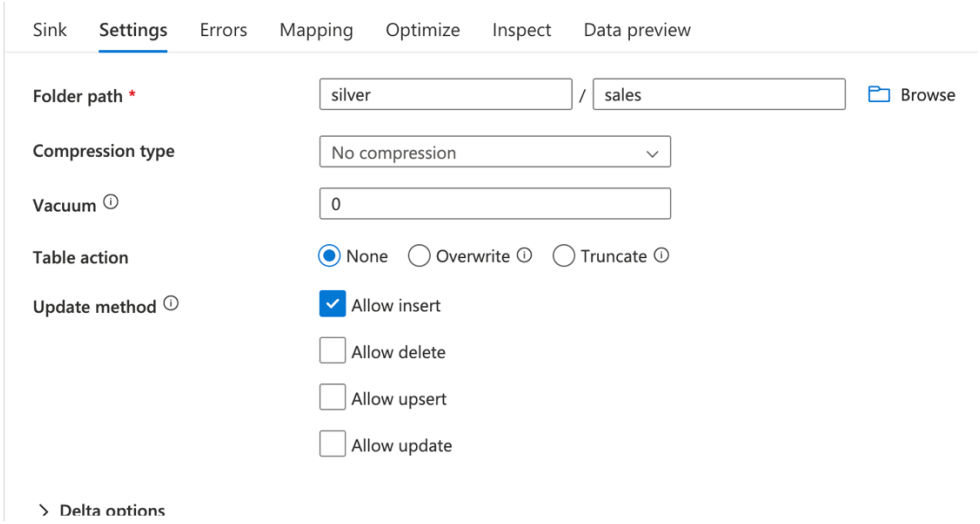
Step 10: Add derived column to add a new column profit



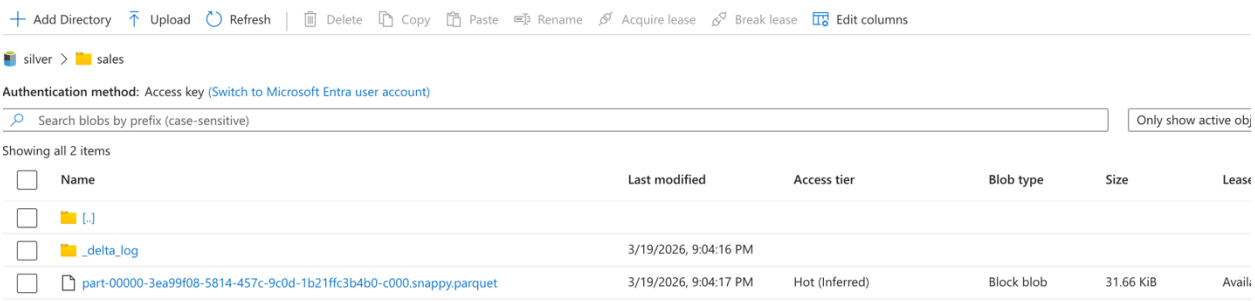

```
round(toDecimal(Unit_Price - Unit_Cost) * Quantity_Sold,2)
```

Step 11: Add a sink activity to save. In sink activity change the sink type to inline, then in inline dataset type select delta to save it as delta table.

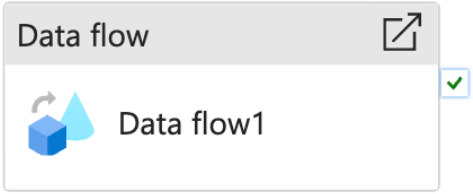




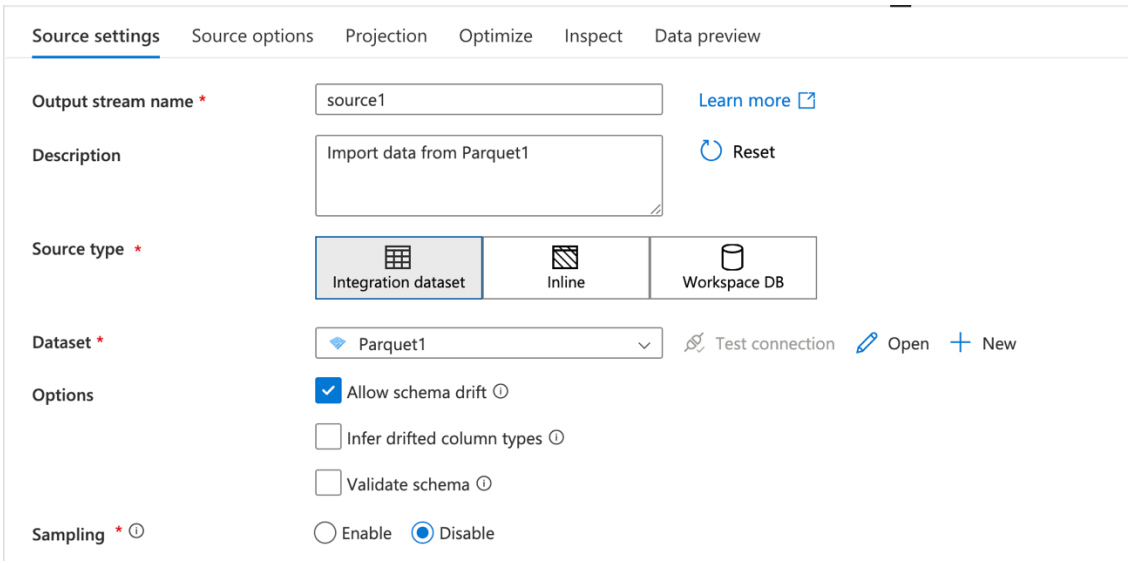
Step 12: Then our first part of the project will be done.



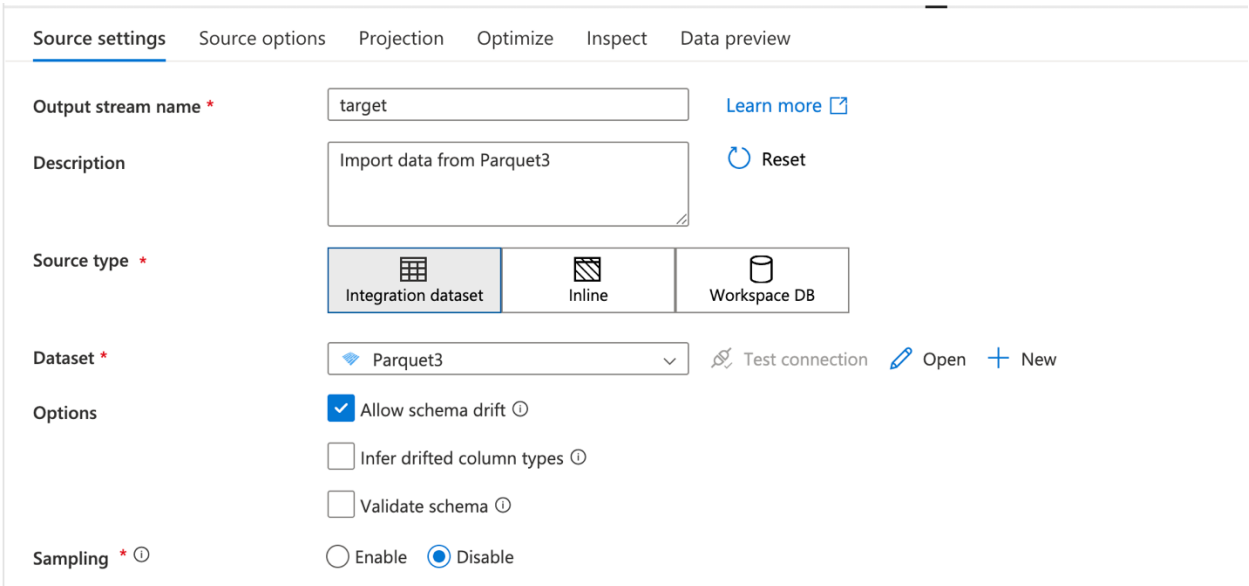
Step 13: Then create one more pipeline and add dataflow similarly to do SCD Type 1



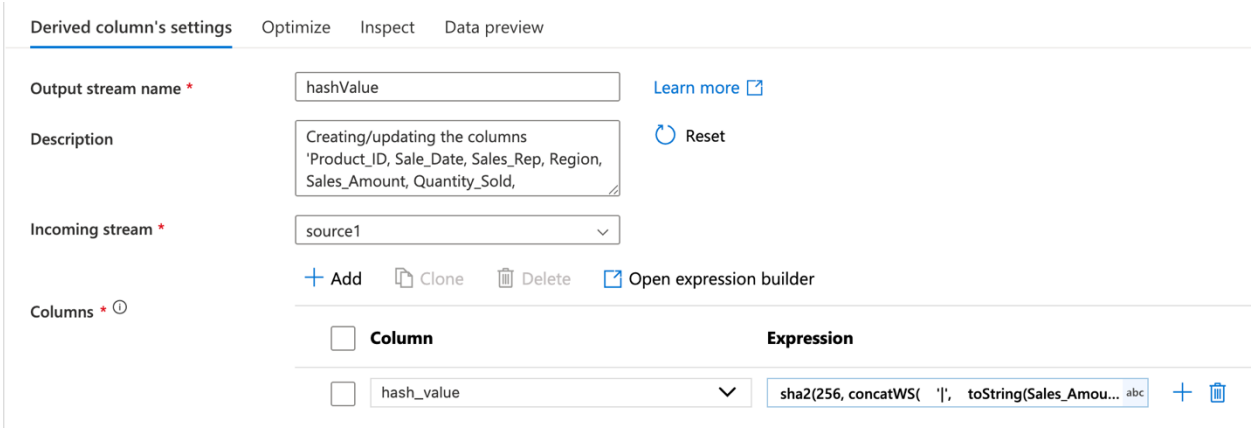
Step 14: In source set it to our silver container delta table that we saved now.



Step 15: Save the same delta table in different location to use it as target to compare to our source.

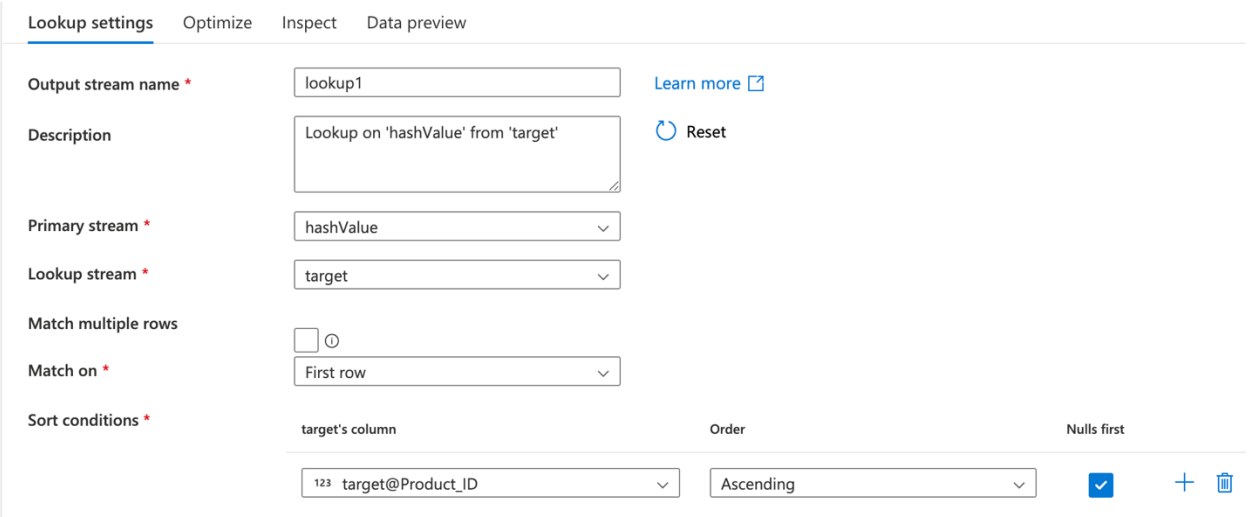


Step 16: Then add a derived column to source to add hash value column.

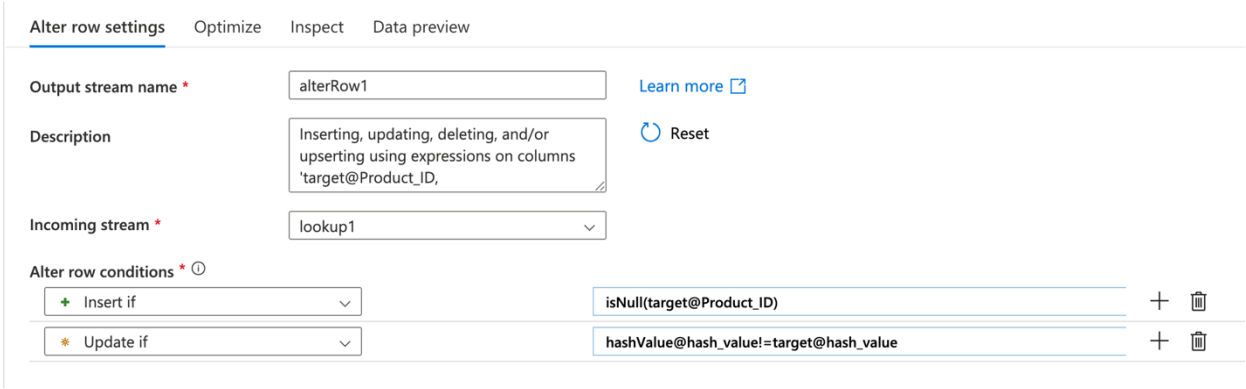



```
sha2(256, concatWS(
    '|',
    toString(Sales_Amount),
    toString(Quantity_Sold),
    toString(Unit_Cost),
    toString(Unit_Price),
    toString(Discount)
  )
)
```

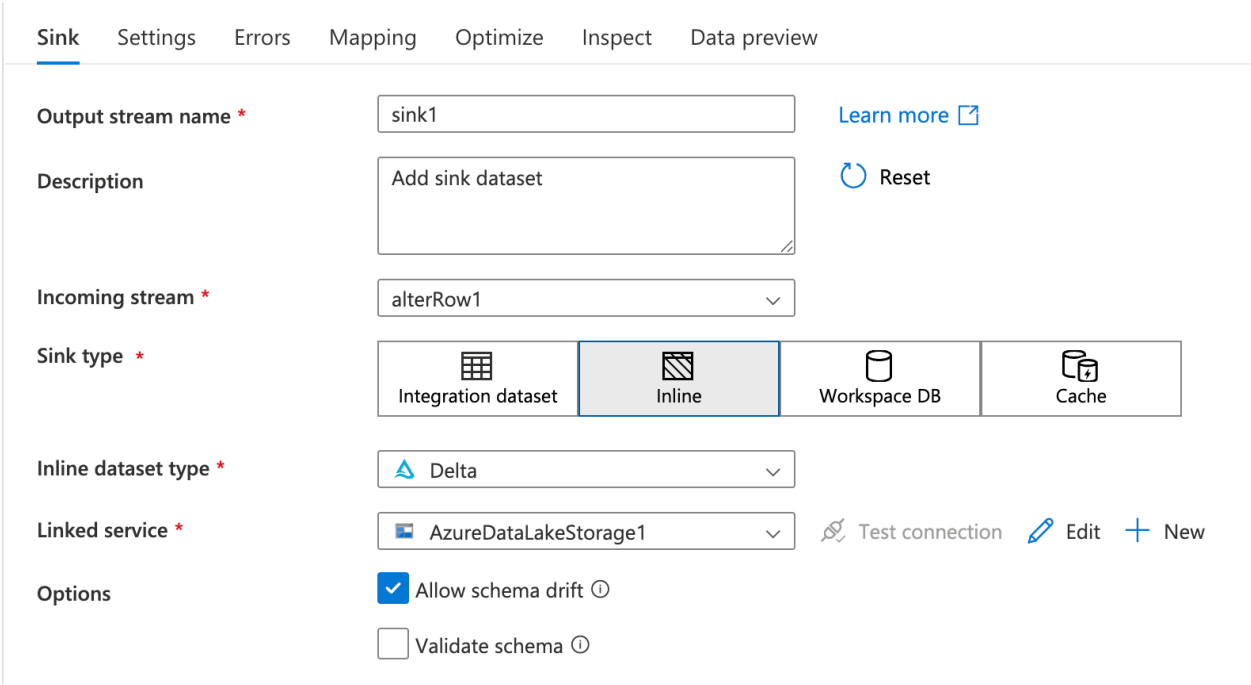
Step 17: Then add a lookup activity to compare and join our source and target.



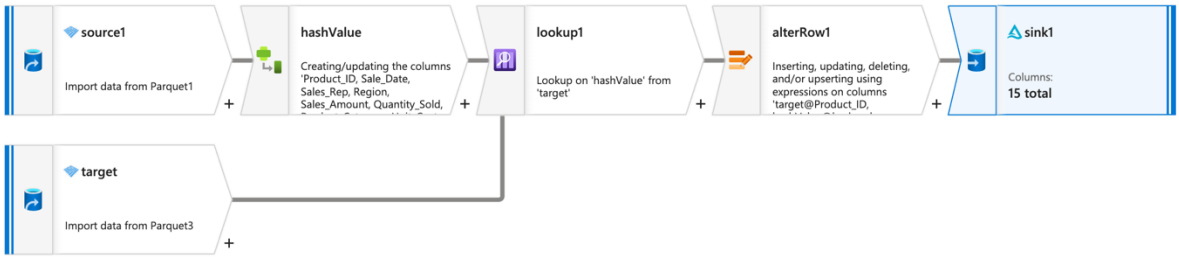
Step 18: Then add a alter row to add our SCD Type 1 condition



Step 19: Then add a sink and save it in delta table format to our gold.



Step 20: Then our second part is also done



Add Directory | Upload | Refresh | Delete | Copy | Paste | Rename | Acquire lease | Break lease | Edit columns

gold > sales

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 2 items

Name	Last modified	Access tier
[..]		
_delta_log	3/22/2026, 11:39:31 PM	
part-00000-00212273-d38c-4efc-bbe5-3c2db638b6c0-c000.snappy.parquet	3/22/2026, 11:39:46 PM	Hot (Inferred)

Step 21: Then go to develop create new SQL script and query your delta table.

Parquet1 | SQL script 2 | Bronze_to_silver | silver_to_gold | Dataflow2

Run | Undo | Publish | Query plan | Connect to Built-in | Use database master

```
SELECT TOP 100 *
FROM OPENROWSET(
    BULK 'https://rithiadls.dfs.core.windows.net/gold/sales/',
    FORMAT = 'DELTA'
) AS rows;
```

Output:

Results | Messages

View: Table | Chart | Export results

Search

Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Categ...	Unit_Cost	Unit_
1001	2023-05-27	Alice	South	1498.11	7	Electronics	4094.68	4576.
1001	2023-11-12	Alice	North	715.81	8	Electronics	3657.91	3729.
1001	2023-08-03	Bob	West	5879.35	20	Food	2153.52	2303.
1001	2023-08-05	Bob	North	8247.54	4	Food	1791.83	1871.
1001	2023-10-08	Bob	West	8397.73	33	Furniture	2933.54	3307.
1001	2023-02-18	David	West	7154.95	27	Clothing	939.02	1286.
1001	2023-04-14	David	East	5488.11	2	Electronics	2631.68	2904.

00:00:46 Query executed successfully.

Create 3 tables with different distribution

- Go to develop, new SQL script and on top click the drop down near connect to
- Select manage pools
- Click new

- Give a name and set performance level to the lowest (DW100c)
- Then review and create this is your dedicated SQL pool

Create a schema

```sql
CREATE SCHEMA project3;
```

```sql
--Table 2 with replicate distribution
CREATE TABLE project3.DimProduct
(
    Product_ID INT NOT NULL,
    Product_Category VARCHAR(50),
    Unit_Cost DECIMAL(10,2),
    Unit_Price DECIMAL(10,2),
    CONSTRAINT PK_DimProduct PRIMARY KEY NONCLUSTERED (Product_ID) NOT ENFORCED
)
WITH
(
    DISTRIBUTION = REPLICATE,
    CLUSTERED COLUMNSTORE INDEX
);
```

```sql
--Table 1 with round-robin distribution
CREATE TABLE project3.DimCustomer
(
    Customer_Key VARCHAR(100) NOT NULL,
    Customer_Type VARCHAR(20),
    Region VARCHAR(20),
    Sales_Channel VARCHAR(20),
    Payment_Method VARCHAR(30),
    Row_Hash VARCHAR(64),
    Load_Date DATE,
    Updated_Date DATE,
    CONSTRAINT PK_DimCustomer PRIMARY KEY NONCLUSTERED (Customer_Key) NOT ENFORCED
)
WITH
(
    DISTRIBUTION = ROUND_ROBIN,
    CLUSTERED COLUMNSTORE INDEX
);
```

```sql
--Table 3 with hash ditribution
CREATE TABLE project3.FactSales
(
    Sale_ID BIGINT NOT NULL,
    Product_ID INT NOT NULL,
    Customer_Key VARCHAR(100) NOT NULL,
    Sale_Date DATE,
    Sales_Rep VARCHAR(50),
    Sales_Amount DECIMAL(18,2),
    Quantity_Sold INT,
    Discount DECIMAL(5,2),
    CONSTRAINT PK_FactSales PRIMARY KEY NONCLUSTERED (Sale_ID) NOT ENFORCED
)
WITH
(
    DISTRIBUTION = HASH(Product_ID),
    CLUSTERED COLUMNSTORE INDEX
);
```

Insert values into the tables

```sql
INSERT INTO project3.DimCustomer
(
    Customer_Key,
    Customer_Type,
    Region,
    Sales_Channel,
    Payment_Method,
    Row_Hash,
    Load_Date,
    Updated_Date
)
VALUES
('NEW_NORTH_ONLINE', 'New', 'North', 'Online', 'Credit Card', 'HASH001', '2023-01-01', '2023-01-01');

SELECT * FROM project3.DimCustomer;
```

Output:

Results Messages

View Table Chart Export results

Search

Customer_Key	Customer_Type	Region	Sales_Channel	Payment_Meth...	Row_Hash	Load_Date	Updated_Date
NEW_NORTH_ONLINE	New	North	Online	Credit Card	HASH001	2023-01-01	2023-01-01

```sql
INSERT INTO project3.DimProduct
(
    Product_ID,
    Product_Category,
    Unit_Cost,
    Unit_Price
)
VALUES
(101, 'Electronics', 500.00, 750.00);

SELECT * FROM project3.DimProduct;
```

Output:

Results Messages

View Table Chart Export results

Search

Product_ID	Product_Category	Unit_Cost	Unit_Price
101	Electronics	500.00	750.00

```sql
INSERT INTO project3.FactSales
(
    Sale_ID,
    Product_ID,
    Customer_Key,
    Sale_Date,
    Sales_Rep,
    Sales_Amount,
    Quantity_Sold,
    Discount
)
VALUES
(1, 101, 'NEW_NORTH_ONLINE', '2023-01-10', 'Alice', 1500.00, 2, 10.00);
SELECT * FROM project3.FactSales;
```

Output:

Results Messages

View Table Chart Export results

Search

Sale_ID	Product_ID	Customer_Key	Sale_Date	Sales_Rep	Sales_Amount	Quantity_Sold	Discount
1	101	NEW_NORTH_...	2023-01-10	Alice	1500.00	2	10.00